

WEEK 2 — BUSINESS UNDERSTANDING REPORT

Customer Churn Prediction for ABC Communications Ltd

Machine Learning Internship Programme

AnalystLab Africa

Week 2 Assignment

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1. Introduction

Customer churn is an important business challenge for telecommunications companies because losing existing customers can reduce revenue and increase the cost of acquiring new customers.

ABC Communications Ltd wants to use machine learning to identify customers who are likely to leave the company. By identifying customers at risk of churn early, the company can take proactive actions such as targeted retention campaigns, personalized offers, improved customer support, and service interventions.

During Week 1, the business problem was investigated and framed as a **binary classification problem** using the Telco Customer Churn dataset.

Week 2 focuses on preparing this data for future machine learning model development. This requires ensuring that the data is clean, consistent, appropriately transformed, and suitable for machine learning algorithms.

2. Company Background

ABC Communications Ltd is a hypothetical telecommunications company that provides services to a large customer base.

Its services include:

- Telephone services
- Internet services
- Online security
- Online backup
- Device protection
- Technical support

- Television streaming
- Movie streaming

The company collects information about customer demographics, service subscriptions, contracts, billing, payment methods, and customer tenure.

This information provides an opportunity to analyze customer behavior and identify characteristics associated with customer churn.

3. Business Problem

- The primary business problem is **customer churn**.
- Customer churn occurs when an existing customer terminates their relationship with the telecommunications provider.
- For ABC Communications Ltd, unexpected customer departures can result in:
 - Loss of recurring revenue.
 - Increased customer acquisition costs.
 - Reduced customer lifetime value.
 - Difficulty maintaining a stable customer base.
 - Reduced effectiveness of long-term customer relationships.

The company therefore needs a data-driven method for identifying customers who are at higher risk of leaving.

The proposed machine learning solution will analyze historical customer information and predict whether a customer is likely to churn.

4. Machine Learning Problem

From a machine learning perspective, the problem can be formulated as a **supervised binary classification problem**.

The model will learn from historical customer records where the actual churn outcome is already known.

The target variable is:

Churn

It contains two possible outcomes:

Value	Meaning
No / 0	Customer did not churn
Yes / 1	Customer churned

The model will eventually learn relationships between customer characteristics and churn outcomes and use those relationships to predict churn for future customers.

5. Dataset Used

The project uses the **Telco Customer Churn Dataset** provided for the AnalystLab Africa internship.

The original dataset contains:

7,043 customer records and 21 columns.

The dataset contains information relating to:

Customer demographics

- Gender
- Senior citizen status
- Partner
- Dependents

Customer relationship

- Tenure

Services

- Phone service

- Multiple lines
- Internet service
- Online security
- Online backup
- Device protection
- Technical support
- Streaming TV
- Streaming movies

Contract and billing

- Contract type
- Paperless billing
- Payment method
- Monthly charges
- Total charges

Target

- Churn

6. Business Questions

The project addresses the following key business questions:

1. Which customers are most likely to churn?

The model will identify customers with characteristics associated with higher churn risk.

2. What customer characteristics are associated with churn?

Features such as tenure, contract type, monthly charges, services, and payment method can be analyzed to identify patterns.

3. Can customer churn be predicted before the customer leaves?

A machine learning model can provide probability or classification indicating whether a customer is likely to churn.

4. How can the company use churn predictions?

ABC Communications Ltd could use predictions to prioritize customers for retention strategies.

7. Target Variable

The target variable is: Churn

It represents whether a customer has left the company.

The original distribution is:

Churn Status	Customers	Percentage
No	5,174	73.46%
Yes	1,869	26.54%
Total	7,043	100%

This demonstrates that the dataset is **imbalanced**, because customers who did not churn substantially outnumber customers who churned.

This imbalance needs to be considered during future model development and evaluation.

8. Input Features

The model will use relevant customer information as input features.

These include:

- gender
- SeniorCitizen
- Partner
- Dependents
- tenure
- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaperlessBilling
- PaymentMethod
- MonthlyCharges
- TotalCharges

Additional engineered features include:

- tenure_group
- total_services

The customerID field is excluded because it is an identifier rather than a meaningful business feature.

9. Why Data Preprocessing Is Important

The Week 2 preprocessing stage was therefore necessary to address several issues.

The preprocessing workflow included:

- Data type correction.
- Missing value treatment.
- Duplicate assessment.
- Removal of irrelevant identifiers.
- Feature engineering.
- Categorical encoding.
- Numerical scaling.
- Outlier detection.
- Feature selection.
- Train-test preparation.

These steps help ensure that the eventual machine learning model receives consistent and meaningful input.

10. Week 2 Preprocessing Connection

The business objective established in Week 1 remains unchanged.

However, Week 2 converts the conceptual machine learning problem into a **machine-learning-ready dataset**.

The preprocessing workflow resulted in:

Original dataset

7,043 × 21



Data cleaning



customerID removed



Feature engineering



Categorical encoding



Numerical scaling



Feature selection



50 processed predictor features

+

1 target

11. Business Benefits

A successful customer churn prediction system could provide ABC Communications Ltd with several business benefits.

11.1 Improved Customer Retention

The company can identify customers who are at risk and intervene before they leave.

11.2 Reduced Revenue Loss

Retaining existing customers can help maintain recurring revenue.

11.3 More Targeted Marketing

Instead of applying the same retention strategy to every customer, the company can focus resources on customers with higher predicted churn risk.

11.4 Better Customer Experience

Churn predictions can help the company identify customers who may require additional support or improved services.

11.5 Data-Driven Decision Making

The system can provide quantitative evidence to support customer retention decisions.

12. Evaluation Considerations

Because churn is a binary classification problem and the dataset contains more non-churn customers than churn customers, accuracy alone should not be used as the only evaluation metric.

Future model development should consider:

- **Precision**
- **Recall**
- **F1-score**
- **ROC-AUC**
- **Confusion Matrix**

Recall will be particularly important because the business may want to identify as many potential churners as possible.

For example, a false negative occurs when the model predicts that a customer will stay when the customer actually churns.

Such an error could cause the company to miss an opportunity to retain that customer.

13. Machine Learning Readiness

Following the Week 2 preprocessing activities, the dataset has been prepared for future model development.

The final dataset has:

Rows: 7,043

Processed predictor features: 50

Target: Churn

Missing values: 0

The data was also divided into:

Dataset	Records	Features
Training	5,634	50
Testing	1,409	50

An 80/20 stratified split was used to preserve the distribution of the churn classes.

14. Expected Future Use

The processed dataset will serve as the foundation for the next stage of the machine learning project.

Future work can include:

1. Training classification algorithms.
2. Comparing model performance.
3. Hyperparameter tuning.
4. Evaluating classification metrics.
5. Selecting the best-performing model.
6. Interpreting important churn predictors.
7. Developing a potential churn-risk prediction system.

Possible algorithms include:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost, if permitted by the project environment

Conclusion

Customer churn represents an important business problem for ABC Communications Ltd because losing customers can negatively affect recurring revenue and increase customer acquisition costs.

The problem has therefore been formulated as a supervised binary classification task in which the objective is to predict whether a customer will churn.

The Week 2 preprocessing stage successfully transformed the original Telco Customer Churn dataset into a machine-learning-ready dataset.

The process addressed data quality and transformation requirements, including data type correction, missing-value treatment, removal of the customerID identifier, feature engineering, categorical encoding, numerical scaling, outlier analysis, and feature selection.

The final dataset contains **7,043 customer records, 50 processed predictor features, and one binary target variable**, with **zero remaining missing values**.

This provides a suitable foundation for the next stage of the project: **machine learning model development, evaluation, and comparison**.